

One House, One Planet



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With the completion of the Intergovernmental Panel on Climate Change (IPCC) synthesis report in November 2007, our mandate is clear. First, we must halt the growth in worldwide green house gas emissions by 2015. Following that, we must sharply reduce emissions, to prevent runaway climate change and the extinction of species. The single largest cause of global warming is coal-fired power plants. The energy produced by coal-fired power plants is used primarily to provide electricity to buildings. In order to stop building new coal-fired plants, and to close existing ones that don't capture carbon, we must improve our buildings by making them more energy efficient, and by powering them with renewable energy.

One Envelope

The two fundamental properties of a well-crafted building envelope are weather seal and insulation. Joints between floors, walls, and ceilings are the seams that need sealing, and the cavities in those areas require insulation. Most conventional homes need air sealing and lack adequate insulation.

Doors and windows need to be weatherstripped and caulked. Other openings—such as recessed-can lights that let air leak to and from the attic—also need to be sealed.

Floors, walls, and attics require insulation. It is relatively easy to insulate an attic, or the floor over a crawlspace, because they're accessible. The walls are another matter. Unless you're replacing siding, insulating walls requires drilling holes to blow or spray insulation into the wall cavities. The

holes are then patched and the walls painted. The paint should be low- or no-VOC (volatile organic compounds) to minimize air pollution.

Wall insulation is a tough sell because of the patching and painting. But remember: We're changing the world, and that takes some resolve. When we insulate the walls, there are no weak links and the reward will be lower utility bills and greater comfort.

One Roof

Most roofs in the United States are asphalt shingles. While warranties on these roofs are for 20 to 50 years, they actually last an average of 15 to 20 years. Asphalt shingles absorb heat from the sun and transfer it to the attic, where temperatures can exceed 130°F on a hot day. It's a losing battle to cool a house with a burning box on top of it, especially if A/C ducts are in that box, struggling to deliver cool air.

The solution is a cool roof. Cool roofs reflect more of the sun's radiation, keep the attic cooler, lessen the need for air conditioning, and reduce the urban heat island effect, whereby air temperatures in urban areas are higher than those in surrounding rural areas. The Cool Roof Rating Council rates cool roofs. If it's not on their Web site, it's not a cool roof. There are two properties that make a roof cool: relatively high solar reflectance (albedo) and relatively high infrared emittance. The former reflects the sun's energy, to minimize the amount of heat the roof absorbs. The latter absorbs heat during the day and releases (reradiates) it at night. One exception is after-market aluminum coatings, which stay warmer at night (and get hotter during the day, since its infrared emittance is low).

Asphalt shingles have a low first cost (although their first cost is rising with the price of oil), but due to their relatively short life span, they have a high lifetime cost. They add about 11 million tons of waste each year to landfills (from tear-offs and

manufacturer scrap). One alternative to asphalt is concrete shingles, shakes, and tile. They are more expensive than asphalt but have longer life spans. Their downside is increased weight, making them unsuitable for most retrofits, and high embodied energy (energy required to manufacture and transport a product to the point of use). The other alternative is metal shingles, shakes, tile, and standing-seam metal. These also have a higher first cost than asphalt, but a longer life span. Metal has a high-recycled content and the lowest embodied energy of the cool-roof choices.

One Window

Windows add character to a home, but what you don't see—the science—is the real story. Low-e coatings, argon or krypton glass fill, and fiberglass composite frames are some of the technologies that are transforming the industry. The right glass and frame for your climate zone, together with the right orientation, will block summer heat, retain winter warmth, and reduce utility costs.

A good source of information on window technologies is the Efficient Windows Collaborative. Their fact sheets (downloadable from their Web site) provide a step-by-step guide to selecting energy-efficient windows for each state. Their interactive Window Selection Tool compares the energy savings of different windows for a typical house.

Retrofitting windows is not always the most cost-effective option, although it is often the first thing on homeowners' minds when they think about retrofitting. Google retrofit windows, and many of the sites that come up will advertise polyvinyl chloride (PVC) replacement windows. A word on PVC: It leaches plasticizers, outgasses additives, and releases poisonous dioxins when it is produced or burned. Even Wal-Mart, Target, Sears, and Kmart are phasing out PVC in products and packaging. There are better choices for windows.

The two options for retrofit windows are to keep the existing frame or to



Native landscaping at my home includes cactus and Mexican Sage.

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A/C came to be considered a necessity only after World War II. Prior to that, we had front porches, wide eaves, and high ceilings. The more time we spent in artificially chilled interiors, the less tolerant we became of temperature and humidity variations, until A/C became an entitlement.

The hotter it gets, the more A/C we use.

The more A/C we use, the hotter it gets. Why? Waste heat from the A/C process contributes to the urban heat island effect, forcing city temperatures up to 10°F hotter than temperatures in rural areas. Their energy use overloads the electrical grid during peak daytime hours, increasing the risk of blackouts. Also, more electricity use equals more global warming.

The best alternative to A/C (and conventional heating) is a GeoExchange (GX) system. For single-family and multifamily homes there should be room to drill vertically, since a GX system uses underground piping to transfer heat to the earth in the summer and extract heat in the winter. The temperature a few feet below the surface is cooler than the outside air in the summer and warmer in the winter, much like the temperature in a cave. According to EPA, GX is the most energy-efficient, environmentally clean, and cost-effective system available, even though the initial cost is high compared to the cost of a conventional system. (Retrofitting with a GX system is much less cost-effective than installing a GX system in new construction.) According to DOE, for a 1,500 ft² home with a good building envelope, GX will heat, cool, and provide hot water for about \$1 a day.

One Renewable Energy System

After improving the energy efficiency of your home and appliances,

replace it. The former is simpler, but it may not be as energy efficient, although it is more cost-effective. Either way requires correct measuring and installation.

One Volt and One Therm

The industry standard for energy-efficient products is Energy Star. Energy Star-labeled products use 10%–50% less energy (and water) than standard models.

Before installing high-efficiency heating and cooling equipment, make sure that the equipment is the proper size, and that it is connected to a properly designed and tight duct system. Consult ACCA *Manual J* for equipment sizing and *Manual D* for designing residential duct systems.

Next on the list—lighting and power strips—are inexpensive. New appliances are more costly, but they're necessary, because old models use an inordinate amount of energy (and water). Replacing electronics can be done when you're in the market for new equipment. Laptop computers use about one-third the power of desktops. With either kind of computer, use power-down or "sleep" mode, and turn the computer off when you're not using it.

One Air Conditioner

Heating and cooling account for almost 50% of the typical American home's annual energy consumption.

and adopting an energy-conserving lifestyle, the final step is installing a renewable energy (RE) system. This comes last because, on average, \$1 spent on energy efficiency saves \$3 to \$5 on the cost of an RE system. Renewable energy options include solar hot water, solar electricity, wind electricity, and microhydro electricity. The solar access of your site, the orientation of your roof, and other design considerations (such as whether your home is historical) all must be taken into account.

One Bird

Improving our homes through energy efficiency and renewable energy will reduce our contribution to global warming. There are other measures we can take to support biodiversity. Two severe consequences of global warming are drought and habitat loss. According to the National Drought Mitigation Center, much of the country—including Alaska and Hawaii—is abnormally dry. Western states are experiencing severe to extreme drought, but the most intense level of drought grips the Southeast.

According to the Audubon Society, one-quarter of U.S. birds are at risk of extinction. One hundred and seventy-six species in the continental United States and 38 in Hawaii are fighting for survival. The causes—drought, invasive species, suburban sprawl, coastal development, industrial farming, and pollution—are widespread. The birds—condors, albatross, ducks, geese, grouse, woodpeckers, and songbirds—are a long list of those we know and love. Birds need to breed, feed, and rest, and our yards can provide them refuge. A Wildlife Habitat certification program, sponsored by the National Wildlife Federation, supports those who want to create a refuge for birds in their yards.

There are a number of ways to transform your home and yard to mitigate drought and habitat loss:

- Use low-flow plumbing fixtures.
- Use composting toilets.

green building

- Use a graywater system for toilet flushing and irrigation.
- Capture rainwater from roof gutters in a filtered barrel or cistern and use it for irrigation and birdbaths.
- Capture rainwater in a vegetated or mulched swale (a landscape area depressed 6 inches) or retention grading (a landscape area over drainage rock) to facilitate ground absorption of rain.
- Capture rainwater in a dry well (a 6-inch-wide grate across a driveway piped to drainage rock under the driveway) to prevent runoff to the street.

These last two measures use soil to filter and clean polluted runoff and recharge the aquifer—our best source of drinking water in a warming world of receding snow pack and high evaporation of reservoirs.

Replace exotic plants with native species for habitat support and carbon storage. Birds and wildlife coevolved with the plants that provide them seeds, fruit, and shelter. Native grasses and plants may be a better choice than trees. Unlike trees, they sequester carbon in the soil and do not drain the aquifer. Plants and trees can also supply shade in areas around the house, reducing cooling energy use, depending on the orientation of the house and the need for south-facing solar access for photovoltaics or passive-solar design.

One Planet

Saving the planet from environmental degradation is a twofold path—what we must do as individuals and what our government must do.

Let's make our representatives hear us, and let's save the planet, one house at a time.

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For more information:

The International Panel on Climate Change (IPCC) report, *IPCC Fourth Assessment Report* is available for download at the IPCC Web site, www.ipcc.ch.

Learn more about cool roofs at www.coolroofs.org.

Learn about recycling asphalt roofing shingles at www.buildinggreen.com.

Learn more about efficient windows at www.efficientwindows.org.

To download a copy of American Architectural Manufacturers Association's *Standard Practice for Installation of Windows with an Exterior Flush Fin over an Existing Window Frame*, AAMA 2410-03, go to www.milgard.com/_doc/products/aama-2410-03.pdf.

For information about the Energy Star program, go to www.energystar.gov.

To find out more about the National Wildlife Federation's Wildlife Habitat certification program, go to www.nwf.org/backyard.